6502 Monitor – ESP32 Pin Mapping (NodeMCU DevKit V1)

Atari ESP32 Project

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Banked RAM PBI / CCTL peripheral (GeoRAM style): 20 bus signals, all through the TXS0108E level translators.

1 Data Bus (Bidirectional)

6502 Signal	ESP32 GPIO	Board Label
D0–D7	GPIO 4, 5, 13, 14, 16, 17, 18, 19	D4, D5, D13, D14, RX2, TX2, D18, D19

2 Address Bus (A0–A7)

6502 Signal	ESP32 GPIO	Board Label
A0–A3	GPIO 34, 35, 36, 39	D34, D35, VP, VN (**Input Only**)
A4–A7	GPIO 32, 33, 21, 27	D32, D33, D21, D27

3 Control Signals

Signal	GPIO	Label	Dir	Description
PHI2	GPIO 2	D2	IN	6502 System Clock (1.79 MHz)
R/W	GPIO 15	D15	IN	Read/Write
∼D1XX	GPIO 22	D22	IN	$D100–$D1FF registers (MMU/PLA, Active Low)
∼CCTL	GPIO 23	D23	IN	$D500–$D5FF RAM window (MMU/PLA, Active Low)

4 Serial Debug

Signal	GPIO	Label	Description
TX	GPIO 1	TX0	Console output (115200 bps, `LOG_BAUD`); also J4 pin 3

5 Unused

GPIO	Label	Note
GPIO 12	D12	U1 / J3 pin 4 (former A8); strapping ignored (VDD_SDIO efuses)
GPIO 25	D25	Free TXS0108E channel (former A9)
GPIO 26	D26	Free TXS0108E channel (former A10)
GPIO 3	RX0	J4 pin 2 only; leave unconnected
GPIO 0	BOOT	Not on the 30-pin headers

6 Commodore 64 (env `c64`, GeoRAM 256 KB)

Same pins; the two select inputs take the C64 I/O lines:

C64 Signal	Expansion Port	J3 Pin	Function
/IO1	pin 7	12 (GPIO 23, ∼CCTL)	RAM window $DE00–$DEFF
/IO2	pin 10	11 (GPIO 22, ∼D1XX)	$DFFE page, $DFFF block (W)

7 Notes

- **A0–A7 only:** the page ($D1 or $D5) comes from the select line.
- **D0–D7** are driven only to answer a read of the RAM window or of a register.
- **Map:** RAM window $D500–$D5FF (current bank); count $D1FB–FC, bank $D1FD–FE (16 bit), PBI latch $D1FF (bit 7).
- U4 channel 8 (pins 9 / 12) and J3 pins 24–25 are unconnected.
- All signals pass through **TXS0108E** bidirectional level shifters (3.3 V ↔ 5 V).